

10 Populations of the moth *Biston betularia* live in Europe and in North America. The most common phenotype on both continents is a pale wing colour with light-grey shading (the typical form).

A moth phenotype with dark wing colour (the melanic form) also occurs on both continents.

Fig. 10.1 shows the typical form of the moth.

Fig. 10.2 shows the melanic form of the moth.



Fig. 10.1



Fig. 10.2

(a) Two melanic European moths were crossed together. The wing colours of the offspring were 15 typical and 41 melanic.

Construct a genetic diagram to explain these results. You may use the symbols **A** and **a** to represent the alleles.

[3]

(b) In a similar experiment, two melanic North American moths were crossed together. The colours of the offspring were 10 typical and 31 melanic.

What can be concluded about the allele that causes the melanic form in the moth populations in **both** continents?

..... [1]

(c) Researchers did not know if the allele causing the melanic form in European moths occurred at the same locus as the allele causing the melanic form in North American moths. To find out, they carried out the following crosses:

- Cross 1: European moths that were heterozygous at the European melanic locus only were crossed with North American moths that were heterozygous at the North American melanic locus only.
- Cross 2: The melanic and the typical offspring of cross 1 were mated together.

(i) Explain why cross 2 is a test cross.

.....

.....

..... [1]

(ii) Complete Table 10.1 to show the predicted results if:

- the European and North American melanic alleles are on **the same locus (A/a)**
- the European and North American melanic alleles are on two **different loci (A/a and B/b)** and **B/b**).

Table 10.1

	same locus (A/a)	different loci (A/a and B/b)
genotypes of melanic moths from cross 1		
proportion of test crosses (cross 2) giving 100% melanic offspring		

[3]

- (d) A light trap was used to estimate the total size of a population of *B. betularia* in a woodland. On night one, 24 moths were captured. These were marked with a small spot of harmless paint. On night two, 29 moths were captured, and 8 of these showed a spot of paint.

Use the Lincoln index formula provided to calculate the size of the population.

Show your working.

$$N = \frac{n_1 \times n_2}{m_2}$$

Key to symbols:

N = estimate of population size

n_1 = number of individuals captured in first sample

n_2 = number of individuals (both marked and unmarked) captured in second sample

m_2 = number of marked individuals recaptured in second sample

population size = [2]

[Total: 10]

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